

Bhavethra Baabu

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EDUCATION

Clark University

Master of Science in Computer Science — GPA: 3.8/4.0

Panimalar Institute of Technology

Bachelor of Technology - Information Technology

Worcester, MA

Aug 2024 – May 2026

Chennai , India

Aug 2018 – Jul 2022

EXPERIENCE

Software Engineer Intern

SciQuel

Jan 2026 – Apr 2026

Allston, MA

- Optimized **PostgreSQL** queries and backend data infrastructure using **Prisma ORM**, reducing metadata retrieval latency by 30% and improving production application performance.
- Implemented **backend services** to retrieve and display leaderboard rankings based on quiz performance, enabling reliable data delivery and enhancing user engagement.
- Automated **build, test, and deployment workflows** using **GitHub Actions**, accelerating production release cycles and reducing manual intervention by 70%.
- Designed **database indexing strategies** for article and profile management services, improving query performance and reducing data retrieval latency by 30%.

Software Engineer

LTIMindtree

Jul 2022 – Jul 2024

Bengaluru, India

- Engineered **Python automation scripts** supporting managed file transfer workflows, improving financial data processing reliability and reducing file transfer failures by 25%.
- Implemented **Python backend services** and **REST API integrations** for enterprise financial systems, reducing transaction processing delays by 25% across more than 40 enterprise clients.
- Developed **Prometheus and Grafana dashboards** tracking 20+ operational KPIs, improving production issue detection time by 35% and strengthening system observability.
- Optimized distributed **data-transfer workflows** for enterprise financial platforms, improving transfer reliability by 25% while supporting scalable backend processing.

PROJECTS

LoadBlast | Python, FastAPI, PostgreSQL, Redis, Docker, REST APIs

May 2026

- Developed a **real-time performance analytics platform** for measuring latency, throughput, and error rates, reducing bottleneck identification time by 60% and improving system observability.
- Implemented **REST APIs** for configuring target endpoints, concurrency levels, ramp-up periods, and test durations, enabling scalable and automated performance testing workflows.

CampusPulse | Next.js, TypeScript, Python, PostgreSQL, REST APIs, Qdrant

Jan 2026 – Apr 2026

- Designed **CampusPulse**, an AI-powered academic search platform leveraging Qdrant and Groq Llama 3.1 to deliver natural language responses across 285+ university pages; presented at the **Clark University Tech Innovation Challenge**.
- Implemented **data ingestion, transformation, and indexing pipelines** to support real-time semantic search and AI-assisted content retrieval through REST APIs.

CoinPulse | Next.js, TypeScript, PostgreSQL, WebSocket, REST APIs

Jan 2026 – Apr 2026

- Constructed a **WebSocket-based real-time data pipeline** integrating Binance APIs to stream live market data and candlestick visualizations for **1,000+ cryptocurrency assets**.
- Optimized **server-side rendering and caching mechanisms**, delivering market updates within **2 seconds** while improving application responsiveness and data delivery performance.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

Frontend: React, Next.js, HTML5, CSS3

Backend: FastAPI, Node.js, REST APIs, Microservices, JSON, JWT, OAuth, WebSockets, HTTP/HTTPS

Databases: PostgreSQL, MongoDB, Redis, Qdrant

Cloud: AWS (EC2, S3, IAM, Lambda, CloudWatch, ECS, EKS), Docker, Kubernetes

DevOps & Tools: Git, GitHub, GitHub Actions, CI/CD, Linux, Jira, ServiceNow

AI/LLM: LangChain, RAG, Vector Search, OpenAI API, Prompt Engineering